On Interpretation and Feature Extraction of Micro-Doppler Response in Reflected Global Navigation Satellite System Signals to Dynamic Sea Surface

Feng Wang, Chuanrui Tan, Weichen Sun, Xiangchao Ma, Faping Lu, Zhichao Xu, Linfeng Wang, Jie Li, Lei Yang, Bo Zhang, and Dongkai Yang

***Abstract*—Conventional coastal global navigation satellite system-reflectometry (GNSS-R) for sea state retrieval relies on the decorrelation of GNSS signals caused by rough sea surfaces, lacking the capacity to capture wave motion. This article demonstrates the detectability of wave motion. Although specular reflection is insensitive to the horizontal component of wave motion, it remains modulated by the vertical wave motion, resulting in a weak time-varying oscillation in the Doppler frequency, referred to as micro-Doppler. A Doppler spectrum, which is related to wave height spectrum, can be derived from the micro-Doppler time series. Besides wave modulation, reflected GNSS signals contain certain low-frequency interferences. To mitigate such interference and reliably extract the micro-Doppler component, a novel method combining dynamic programming and empirical mode decomposition (EMD) is proposed. Some characteristics of the micro-Doppler are defined to quantize its dependencies on wind speed, wind direction, wave direction, swell, and mean wave period. A significant finding lies in the existence of off-specular scattering in GNSS-R forward geometry, thereby establishing a physical basis for wind direction retrieval. The experimentally obtained relationships between these characteristics and ocean parameters agree well with theoretical predictions, confirming the modulation of wave motion on reflected GNSS signals. The wave height spectrum is derived from the Doppler spectrum. Wind speed, wind direction, significant wave height (SWH), and mean wave period are retrieved, with root mean square errors (RMSEs) of 2.00 m/s, 32.0°, 0.50 m, and 1.5 s, respectively.**

***Index Terms*—Coastal global navigation satellite system-reflectometry (GNSS-R), Doppler spectrum, sea wave spectrum, significant wave height (SWH), wave motion, wave period, wind speed.**

I. Introduction

COASTAL ocean physical parameters, including wind speed, significant wave height (SWH), and mean wave period, are critical for coastal engineering, hazard monitoring, and oceanographic studies. Buoys, serving as in situ sensors, have been instrumental in providing these data. However, capturing their spatial variability typically requires a dense network of buoys, which entails significant maintenance and financial challenges. Microwave remote sensing technologies, such as scatterometers [1], altimeters [2], and synthetic aperture radar (SAR) [3], have been employed to estimate these parameters from space, providing large-scale coverage and proven effectiveness in the open ocean. Yet their applications in coastal regions remain limited due to coarse temporal resolutions, high costs, and high power consumption, which also make them unsuitable for deployment on coastal platforms. Consequently, developing a novel remote sensing solution with low cost and low power consumption is essential for coastal monitoring. Signals-of-Opportunity (SoOp) remote sensing offers a promising alternative by leveraging reflections from existing illuminators, such as communication and navigation satellites, to retrieve Earth's physical parameters. A prominent example is global navigation satellite system-reflectometry (GNSS-R). Because it does not require a dedicated transmitter, SoOp remote sensing holds significant cost and power advantages over these conventional active microwave systems. This technology has emerged as a viable means for measuring wind speed, SWH, and soil moisture from both ground-based and spaceborne platforms [4], [5], [6], [7], [8]. In coastal regions, the coherence time of reflected GNSS signals has been demonstrated to be sensitive to sea state [9], leading to the development of dedicated real-time instruments for monitoring sea state [10], [11]. The methodology has since been extended

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to incorporate S-band and Ku-band reflected signals [12] [13], and frequency-domain approaches based on spectral width have also been established for sea state retrieval [14], [15]. To maximize the signal-to-noise ratio (SNR), existing GNSS-R systems have primarily focused on specular reflections. This observation geometry differs fundamentally from the backscattering mechanisms utilized by active radars, which are effective for detecting wave motions, such as the orbital velocity of wave particles along the radar beam. This phenomenon has been successfully applied to measure wave height spectrum [16], [17]. In contrast, because the sum of the projections of horizontal velocity onto the incident and reflected directions is zero, specular reflection is largely insensitive to horizontal motion; consequently, few works have explored wave motion measurement using coastal GNSS-R. The first objective of this article is to address this gap. The presence of a micro-Doppler phenomenon in specular reflection induced by wave motion is theoretically and experimentally demonstrated. The second objective is to investigate the potential for retrieving the wave height spectrum from this micro-Doppler signal, and to propose a novel method for estimating SWH and mean wave period using specular reflection. Given the extensive availability of experimental data, GNSS-R is employed as a representative case study in this article. It is anticipated that the methodologies and findings presented here can be extended to other SoOp remote sensing applications. The remainder of this article is organized as follows. Section II presents the theoretical foundations of GNSS-R, including its geometry and signal model, and analyzes the micro-Doppler effect in reflected GNSS signals from a dynamic sea surface, derives the corresponding Doppler spectrum theoretically, and defines key micro-Doppler characteristics. Section III details the methods for extracting the micro-Doppler and its characteristics. Experimental validation and analysis of the micro-Doppler characteristics are provided in Section IV. Section V explores the application of the micro-Doppler phenomenon to sea state observation. Sections VI and VII discuss and conclude this article.

II. Micro-Doppler Effect in Forward Scattered GNSS Signals

A. GNSS-R Geometry

GNSS-R is a forward-scattering observation technique that utilizes specularly reflected GNSS signals for sea surface monitoring. The observation geometry is illustrated in Fig. 1. For a forward geometry, the vectors of incident and scattered GNSS signals are expressed as

$$\begin{cases} \mathbf{k}_i = \dfrac{2\pi}{\lambda}\left(\cos\theta_i\cos\varphi_i, \cos\theta_i\sin\varphi_i, -\sin\theta_i\right) \\ \mathbf{k}_r = \dfrac{2\pi}{\lambda}\left(\cos\theta_r\cos\varphi_r, \cos\theta_r\sin\varphi_r, \sin\theta_r\right) \end{cases} \tag{1}$$

where θ_i and θ_r are, respectively, the elevation angles of the incident and scattered signals; φ_i and φ_r are, respectively, the azimuth angles of the incident and scattered signals; and λ is the wavelength of the GNSS signal. The scattering vector $\mathbf{q}$ is defined as $\mathbf{k}_r - \mathbf{k}_i$. In a specular geometry, the elevation angle of the scattered signal equals that of the incidence signal

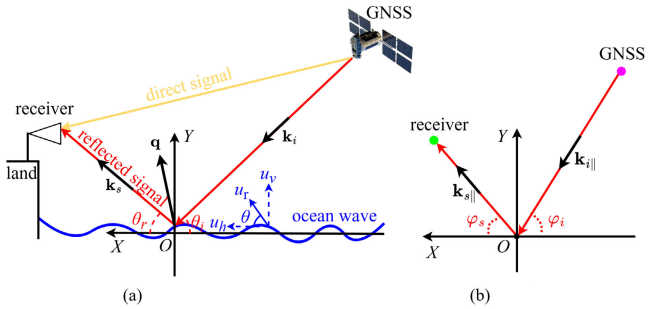


Fig. 1. Geometry of reflection of the GNSS signal from the sea surface. $\mathbf{k}_{i\|}$ and $\mathbf{k}_{s\|}$ denote, respectively, the horizontal components of $\mathbf{k_i}$ and $\mathbf{k_s}$. (a) Front view. (b) Top view.

($\theta_i = \theta_r$), and the azimuth angles satisfy the relation $\varphi_r = \varphi_i$. For a receiver at a height h_r, the observation region of specular geometry is centered around the specular point, whose coordinates are given by

$$\begin{cases} e = \dfrac{h_\mathrm{r}}{\tan\theta_i}\cos\varphi_i \\ n = \dfrac{h_\mathrm{r}}{\tan\theta_i}\sin\varphi_i. \end{cases} \tag{2}$$

B. Signal Model

Assuming unit amplitude and a single reflection echo, the reflected signal can be expressed as

$$s_r(t) = \exp\{-j2\pi f(t - \tau(t))\} \tag{3}$$

where f is the carrier frequency, and $\tau(t)$ is the time-varying delay between the direct and reflected GNSS signals, computed as

$$\tau(t) = \frac{2(h_\mathrm{r} + \xi(x,t))\sin\theta}{c} \tag{4}$$

where $\xi(x,t)$ represents the dynamic sea surface, and c is the speed of light. Modeling the sea surface as a single long ocean wave yields

$$\xi(x,t) = A\cos(kx \pm \omega t) \tag{5}$$

where A, k, and ω are the amplitude, wavenumber, and angular frequency of the sea wave, respectively. The angular frequency and wavenumber satisfy the dispersion relation as

$$\omega = \sqrt{gk\tanh(kd)} \tag{6}$$

where g is the gravitational acceleration; and d is water depth. The phase variation induced by the delay, derived from (4) to (5), is

$$\phi_p(t) = 2\pi f\frac{\tau(t)}{c} = \frac{4\pi(h_\mathrm{r} + A\cos(kx \pm \omega t))\sin\theta}{\lambda} \tag{7}$$

where λ is the GNSS signal wavelength. Assuming a constant elevation angle over a short observation interval, the instantaneous Doppler frequency is derived from this phase as

$$f_\mathrm{Dopp}(t) = \frac{1}{2\pi}\frac{d\phi_p(t)}{dt} = \frac{2A\omega\sin(kx \pm \omega t)\sin\theta}{\lambda}. \tag{8}$$

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C. Wave Motion in the Sea Surface

The time-varying Doppler can also be interpreted through the orbital motion of sea waves. According to linear wave theory, the horizontal and vertical orbital velocity components for a single long wave are expressed as [18]

$$\begin{cases} u_h(x,t) = A\omega \coth(kd)\cos(kx \pm \omega t) \\ u_z(x,t) = A\omega \sin(kx \pm \omega t). \end{cases} \tag{9}$$

In a bistatic observation, the Doppler frequency caused by this orbital velocity is expressed as

$$f_{\text{Dopp}}(t) = \frac{1}{\lambda}(\mathbf{v}\cdot(\mathbf{k}_r - \mathbf{k}_i)) = \frac{v_d}{\lambda} \tag{10}$$

where v_d is the relative velocity component along the line-of-sight, given by

$$v_d = v_h(\cos\theta_r - \cos\theta_i) + v_z(\sin\theta_r + \sin\theta_i) \tag{11}$$

where θ_i and θ_r are the included angles between the horizontal direction and incident and reflected wave vectors, respectively. Under specular reflection, the line-of-sight velocity is simplified to $v_d = 2v_z\sin\theta$, leading to a Doppler frequency of

$$f_{\text{Dopp}}(t) = \frac{2A\omega\sin(kx \pm \omega t)\sin\theta}{\lambda}. \tag{12}$$

Equations (8) and (12) are identical, confirming the consistency of the two analytical approaches.

D. Micro-Doppler Phenomenon

In the coastal GNSS-R scenario, the bulk motion of the GNSS satellite induces a dominant, constant Doppler shift in the reflected GNSS signal, superimposed on a weak and time-varying Doppler modulation arising from ocean wave orbital motion. This dynamic component, which is significantly smaller in magnitude, is termed the micro-Doppler effect. As illustrated in (12), for a single long-period wave, the micro-Doppler frequency oscillates at the wave frequency itself. Fig. 2 presents time-Doppler spectra of reflected GNSS signals under wind speeds of 8.6 and 2.1 m/s. The constant Doppler component has been removed. The time-varying fluctuations in the Doppler peak, induced by the motions of ocean waves, are clearly visible. The spectrum under high wind speed exhibits both a larger fluctuation amplitude and a longer periodicity in the Doppler peaks compared to the low-wind-speed case. These observations align with the expectation that stronger winds generate ocean waves with greater amplitudes and longer periods.

In reality, the sea surface is a superposition of many waves with different frequencies and amplitudes. The total micro-Doppler modulation is therefore the summation of contributions from all constituent waves

$$f_{\text{Dopp}}(t) = \frac{2\sin\theta}{\lambda}\sum_i A(\omega_i)\omega_i \sin(k_i x \pm \omega_i t) \tag{13}$$

where k_i and ω_i are the wavenumber and angular frequency of the ith ocean wave, respectively.

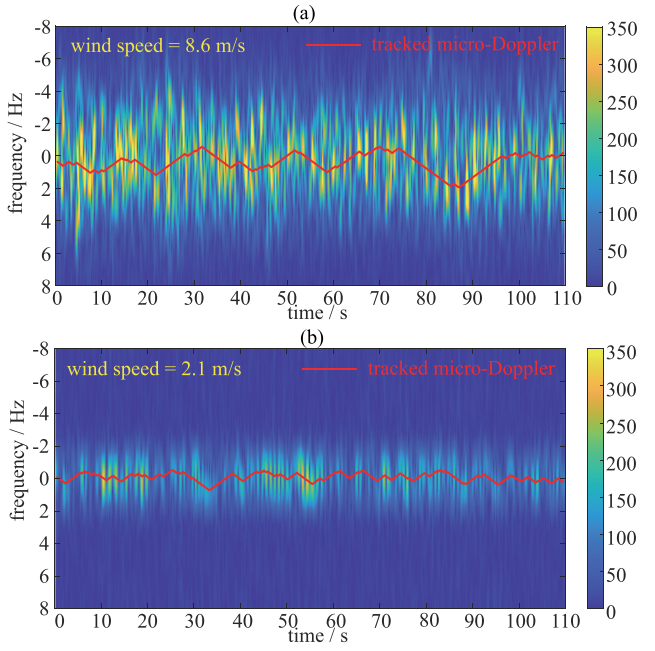


Fig. 2. Time-Doppler spectra of the reflected GNSS signal at wind speeds of (a) 8.6 m/s and (b) 2.1 m/s. The red lines are the estimated Doppler peaks.

E. Doppler Spectrum

The Doppler spectrum is defined as the power spectral density of the time-varying Doppler

$$F(f) = \left|\text{FFT}\left[f_{\text{Dopp}}(t)\right]\right|^2. \tag{14}$$

According to (12), the Doppler spectrum of a single long wave is a Dirac delta function

$$F(f) = \frac{4A^2(\omega_i)\omega_i^2\sin^2\theta}{\lambda^2}\delta\left(f - \frac{\omega_i}{2\pi}\right) \tag{15}$$

where $\delta(f)$ is the Dirac function. For a sea surface comprising multiple waves, the total Doppler spectrum integrates individual contributions and is expressed as

$$F(f) = \frac{4\sin^2\theta}{\lambda^2}\sum_i \omega_i^2 A^2(\omega_i)\,\delta(2\pi f - \omega_i). \tag{16}$$

In continuous form, this becomes

$$F(f) = \frac{16\pi^2\sin^2\theta}{\lambda^2} f^2 A^2(f). \tag{17}$$

The wave amplitude is related to the wave spectrum as

$$\frac{1}{2}A^2(f) = S(f)\cdot f_\Delta \tag{18}$$

where $S(f)$ is the sea wave spectrum, and f_Δ denotes the frequency resolution. Substituting (18) into (17), and accounting for wave directionality, the Doppler spectrum is expressed as

$$F(f) = \frac{32\pi^2\sin^2\theta f_\Delta f^2}{\lambda^2} S(f,\varphi) \tag{19}$$

where φ is the included angle between wind and wave directions defined as $\varphi_{\text{wind}} - \varphi_{\text{wave}}$. A directional wave spectrum can

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be factorized into a nondirectional frequency spectrum and a directional spreading function as

$$S(f,\varphi) = S_a(f) \cdot D(f,\varphi) \tag{20}$$

where $S_a(f)$ is the nondirectional frequency spectrum, and $D(f,\varphi)$ is the directional spreading function. A commonly used parametric spreading function is cosine-2s model as [19]

$$D(f,\varphi) = \cos^{2s}\left(\frac{\varphi}{2}\right) \tag{21}$$

where s is the spreading parameter, dependent on wavenumber, wind speed, and inverse wave age. The final expression for the Doppler spectrum is given as

$$F(f) = \frac{32\pi^2 \sin^2\theta f_\Delta f^2}{\lambda^2} S_a(f) \cos^{2s}\left(\frac{\varphi}{2}\right). \tag{22}$$

In this article, the Pierson–Moskowitz spectrum [20] is adopted for the nondirectional frequency spectrum.

F. Micro-Doppler Characteristics

To quantify the micro-Doppler, several characteristic metrics are defined.

1) Doppler Spectrum Characteristics: The first characteristic is the peak frequency of the Doppler spectrum, at which the Doppler spectrum attains its maximum value

$$f_{\mathrm{p}} = \arg\max_f \{F(f)\}. \tag{23}$$

The second characteristic is the averaged frequency of the Doppler spectrum, which is evaluated using the spectrum moment method as

$$f_{\mathrm{a}} = \frac{\int_0^{+\infty} f \cdot F(f)\, df}{\int_0^{+\infty} F(f)\, df}. \tag{24}$$

The third feature is the Doppler spectral peak as

$$S_{\mathrm{peak}} = \max\{F(f)\}. \tag{25}$$

The fourth characteristic is the Doppler spectral width as

$$B_{\mathrm{S}} = \frac{\int_0^{+\infty} (f - f_{\mathrm{a}})^2 \cdot F(f)\, df}{\int_0^{+\infty} F(f)\, df}. \tag{26}$$

2) Micro-Doppler Amplitude: Two micro-Doppler amplitudes can be derived from the micro-Doppler series. The maximum amplitude of the Doppler series is expressed as

$$A_{\mathrm{fmax}} = \max\left\{f_{\mathrm{Dopp}}(t)\right\}. \tag{27}$$

From (22), the maximum amplitude is obtained from the Doppler spectrum peak and its corresponding frequency as

$$A_{\mathrm{fmax}} = \frac{4\pi\sqrt{2}\sin\theta f_{\mathrm{peak}}\sqrt{S_{\mathrm{peak}} f_\Delta}}{\lambda} |\cos^s\left(\frac{\varphi}{2}\right)|. \tag{28}$$

Another term is the effective value or the root mean square (rms) of the amplitude, defined as

$$\sigma_f = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N-1}\left(f_{\mathrm{Dopp}}(t_i) - \overline{f}_d\right)^2} \tag{29}$$

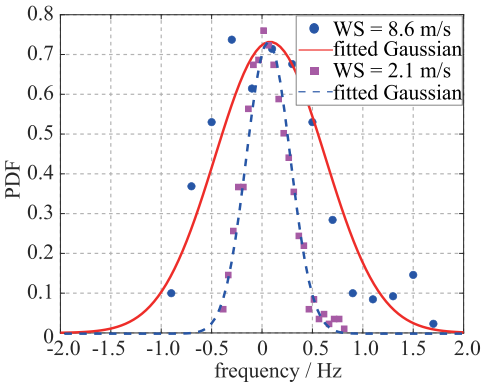


Fig. 3. Distributions of the micro-Doppler series and corresponding fit Gaussian functions at wind speeds of 8.6 and 2.1 m/s.

where N is the sample number of the micro-Doppler frequency, and $\overline{f}_d$ is the mean micro-Doppler frequency shift, defined as

$$\overline{f}_d = \langle f_d(t) \rangle. \tag{30}$$

For a square wave, the ratio of the maximum amplitude to the rms is 1; for a sine wave, this ratio is 1.42; for a triangle wave and for a time series following a Gaussian distribution, the ratios are 1.73 and approximately 3–4, respectively [21]. According to the Central Limit Theorem (CLT), the sum of numerous independent wave contributions in (13) results in a Gaussian distribution of amplitude (see Fig. 3). From the Gaussian distribution of the Doppler series, the second amplitude is defined as $A_{\mathrm{eff}} = \sigma_{\mathrm{f}}$, where σ_{f} is the standard deviation of the Gaussian distribution of the Doppler series.

3) Mean Micro-Doppler Frequency Shift: According to (13), the mean micro-Doppler frequency equals zero when off-specular scattering and selective scattering of ocean waves are neglected. In a forward geometry and under the geometrical-optics approximation, the averaged Doppler frequency shift of the scattered signal is obtained by statistically averaging the velocities of all reflection points that satisfy the specular condition

$$\bar{f}_d = -\frac{\mathbf{q}}{2\pi} \cdot \left\langle \frac{\partial \xi(\mathbf{r},t)}{\partial t} \right\rangle_{\nabla\xi(\mathbf{r},t)=\mathbf{s}} \tag{31}$$

where $\langle\cdot\rangle|_{\nabla=\mathbf{s}}$ denotes the conditional expectation given $\nabla = \mathbf{s}$, $\nabla\xi$ is the spatial gradient of the sea surface, and $\mathbf{s}$ represents the gradient vector for specular scattering, defined as

$$\mathbf{s} = \frac{-\mathbf{q}_\perp}{q_z} = -\left(\frac{q_x}{q_z}, \frac{q_y}{q_z}\right)^T \tag{32}$$

where $\mathbf{q} = (q_x, q_y, q_z)$. Assuming the sea surface is a zero-mean Gaussian random process, its spatial gradient and time derivative are also Gaussian distributed. The conditional expectation in (31) can therefore be evaluated as

$$\begin{aligned}\left\langle \frac{\partial \xi(\mathbf{r},t)}{\partial t} \right\rangle_{\nabla\xi(\mathbf{r},t)=\mathbf{s}} &= \left(\langle q_{tx}\rangle, \langle q_{ty}\rangle\right) \cdot \begin{pmatrix} \langle q_x^2\rangle & 0 \\ 0 & \langle q_y^2\rangle \end{pmatrix}^{-1} \cdot \mathbf{s} \\ &= \left(-\frac{\langle q_{tx}\rangle}{\langle q_x^2\rangle}\frac{q_x}{q_z}, -\frac{\langle q_{ty}\rangle}{\langle q_y^2\rangle}\frac{q_y}{q_z}\right)^T \end{aligned} \tag{33}$$

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where $\langle q_x^2 \rangle$ and $\langle q_y^2 \rangle$ are the directional mean-square slopes, respectively; $\langle q_{tx} \rangle$ and $\langle q_{ty} \rangle$ are the respective spatial–temporal cross derivatives. Note that the cross-term of the mean-square slope $\langle q_x q_y \rangle$ is taken to be zero. The directional mean-square slopes and spatial–temporal cross derivatives are computed from the wave height spectrum as

$$\begin{cases} \langle q_x^2 \rangle = \int_0^{+\infty} \int_0^{2\pi} k^3 \cos^2\varphi S\,(k,\varphi)\, d\varphi dk \\ \langle q_y^2 \rangle = \int_0^{+\infty} \int_0^{2\pi} k^3 \sin^2\varphi S\,(k,\varphi)\, d\varphi dk \\ \langle q_{tx} \rangle = \int_0^{+\infty} \int_0^{2\pi} \omega k^2 \cos\varphi S\,(k,\varphi)\, d\varphi dk \\ \langle q_{ty} \rangle = \int_0^{+\infty} \int_0^{2\pi} \omega k^2 \sin\varphi S\,(k,\varphi)\, d\varphi dk \end{cases} \tag{34}$$

where $S(k,\varphi)$ denotes the wavenumber spectrum of the sea wave. Consequently, the mean micro-Doppler frequency shift can be expressed as a function of the directional mean-square slopes, the spatial–temporal cross derivatives, and the scattering vector

$$\overline{f}_d = \frac{1}{q_z}\left(\frac{\langle q_{tx} \rangle}{\langle q_x^2 \rangle} q_x^2 + \frac{\langle q_{ty} \rangle}{\langle q_y^2 \rangle} q_y^2\right). \tag{35}$$

For an in-plane geometry $\varphi_r = \varphi_i$ and an observation direction along the X-axis, the mean micro-Doppler frequency shift simplifies to

$$\overline{f}_d = \frac{\sin\theta_i \sin^2\Delta\theta}{2\lambda} \frac{\langle q_{tx} \rangle}{\langle q_x^2 \rangle} \tag{36}$$

where $\Delta\theta = \theta_r - \theta_i$. For pure specular reflection with $\Delta\theta = 0$, the mean micro-Doppler frequency shift vanishes. For nonzero $\Delta\theta$, the resulting mean frequency shift becomes sensitive to the directional mean-square slopes, suggesting the feasibility of retrieving wind direction. Notably, only under local wind forcing do sea waves propagate in a direction aligned with the wind direction. However, when swell dominates, the wave direction may deviate significantly from the wind direction, which can compromise wind direction retrieval and lead to a reduction in retrieval accuracy. Fig. 4 illustrates the mean micro-Doppler frequency shift for different wind speeds and directions with $\Delta\theta = 2°$. The shift varies symmetrically with wind direction about a 180° axis, implying a 180° ambiguity when used for wind direction retrieval. As wind speed increases, the amplitude of the mean micro-Doppler frequency shift exhibits a corresponding growth.

III. Micro-Doppler and Feature Extraction Approaches

Building upon the theoretical foundation established in Section II, an methodology for extracting the micro-Doppler time series and its corresponding Doppler spectrum is proposed. The detailed flowchart of the procedure is shown in Fig. 5. The primary input observable to the algorithm is the normalized complex amplitude series of the reflected GNSS signal, defined as

$$icf(t) = \frac{I_r(t) + jQ_r(t)}{I_d(t) + jQ_d(t)} \tag{37}$$

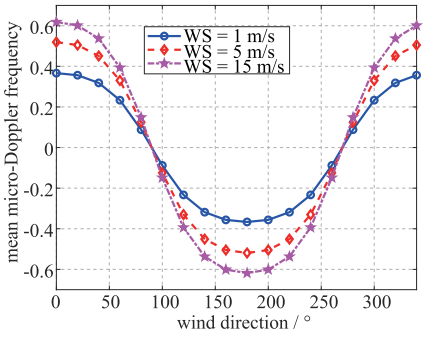


Fig. 4. Mean micro-Doppler frequency shift for different wind speeds and directions ($\Delta\theta = 2°$).

where I_r and Q_r are the in-phase and quadrature correlation values of the reflected GNSS signals, respectively; I_d and Q_d are the corresponding correlation values from the direct GNSS signals. This normalization by the direct signals mitigates the influence of common-mode factors, such as navigation message and transmit power variations. The complex correlation values are sampled at 1000 Hz.

1) Estimation of Time-Doppler Spectrum:

The first step involves computing the short-time Fourier transform (STFT) of the normalized complex correlation values to obtain a time-Doppler spectrogram. A Hamming window of length 640 samples is employed to minimize spectral leakage. To compensate for attenuation at the segment edges caused by the Hamming window, an overlap of 90% is used between successive segments, ensuring no data is lost. The resulting temporal resolution of the time-Doppler spectrogram is 64 ms.

2) Tracking of Micro-Doppler Series:

The micro-Doppler series is extracted by tracking the spectral ridge across the time-Doppler spectrogram. Given the inherently low signal-to-noise ratio of the reflected GNSS signal, a dynamic programming algorithm is employed for robust ridge extraction. The core state transition equation of dynamic programming is

$$V(t,f) = S(t,f) + \max_{f' \in N(f)} \left\{ V\left(t-1, f'\right) \right\} \tag{38}$$

where $S(t,f)$ is the time-Doppler spectrogram intensity, and $N(f)$ defines a frequency neighborhood around f

$$N(f) = \left\{ f' : |f' - f| \le D \right\} \tag{39}$$

where D is a preset maximum frequency transition threshold. After populating the cumulative cost matrix $V(T,f)$, the optimal terminal frequency at time T is selected as

$$\widehat{f}_{\text{Dopp}}(T) = \arg\max_f \left[V(T,f) \right]. \tag{40}$$

The complete micro-Doppler series is obtained through the backtracking formula as

$$\widehat{f}_{\text{Dopp}}(t-1) = \arg\max_{f' \in N\left(\widehat{f}(t)\right)} \left[V\left(t-1, f'\right) \right]. \tag{41}$$

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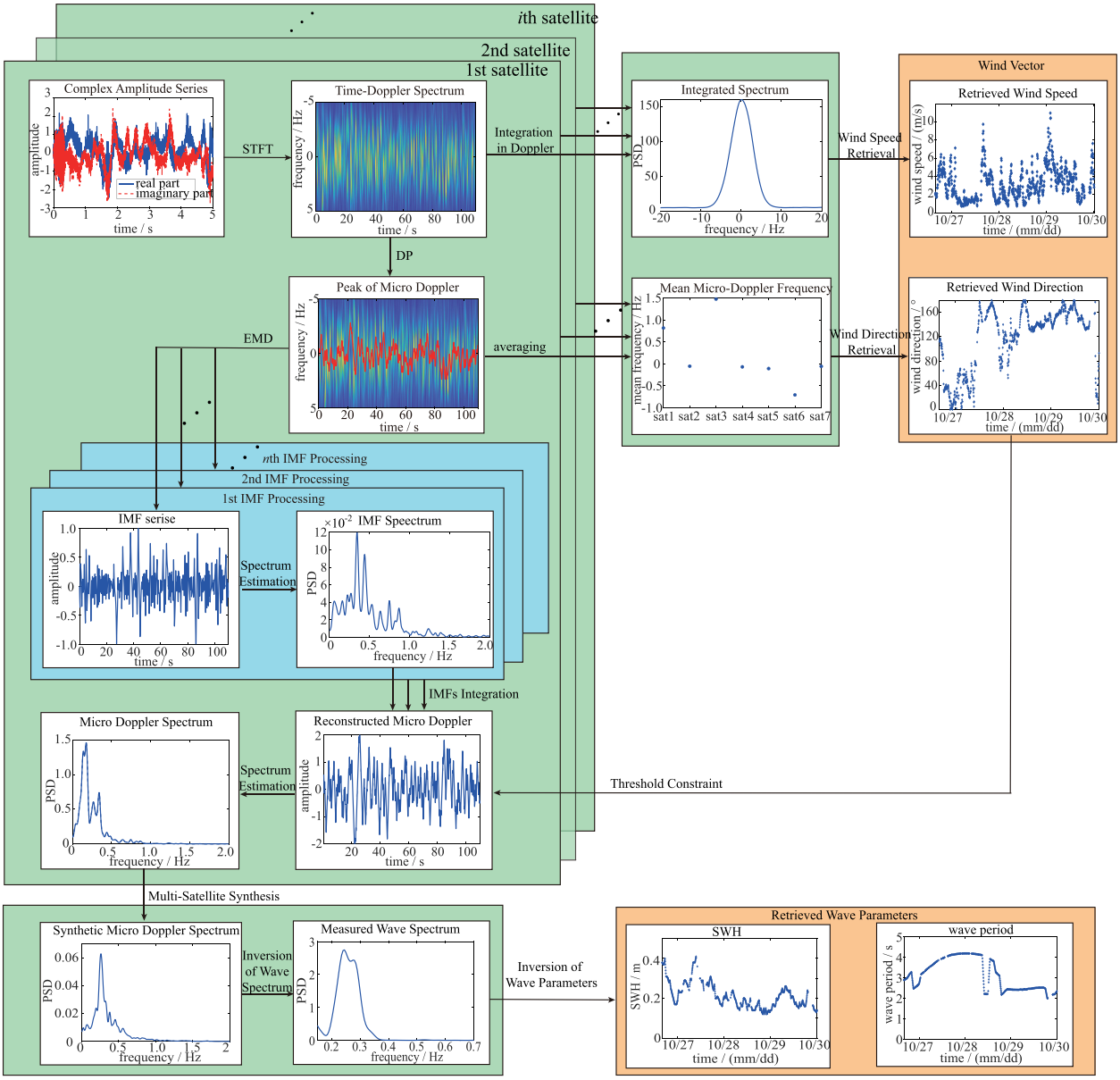


Fig. 5. Flowchart of the proposed approach for extracting the wave spectrum.

The red curves in Fig. 2(a) and (b) present the tracked micro-Doppler series using dynamic programming for wind speeds of 8.6 and 2.1 m/s, respectively.

3) Estimation of Doppler Spectrum:

The Doppler spectrum is obtained via the Fourier transform of the extracted micro-Doppler series, as defined in (14). To reduce the estimation variance, Welch's method is applied. Fig. 6(a) and (c) presents the estimated Doppler spectra for wind speeds of 8.6 and 2.1 m/s, respectively, alongside their theoretical counterparts (red curves). A notable feature of the estimated Doppler spectra is the presence of strong low-frequency components (within the black dashed boxes), likely attributable to nonmotion factors, such as wave breaking [22]. These low-frequency components can cause significant overestimation of the wave spectrum at low frequencies, and subsequently bias derived wave parameters like the mean wave period.

4) Extraction of Wave-Motion-Induced Micro-Doppler:

From Fig. 6(a) and (c), it is seen that the Doppler spectrums induced by these nonmotion factors are mainly low-frequency components, while those induced by wave motions are distributed in the high-frequency region. To mitigate the micro-Doppler component induced by nonmotion factors, empirical mode decomposition (EMD) is employed. EMD adaptively decomposes the nonlinear and nonstationary series into a set of intrinsic mode functions (IMFs) as

$$\widehat{f}(t) = \sum_{i=1}^{N} c_i(t) + r(t) \tag{42}$$

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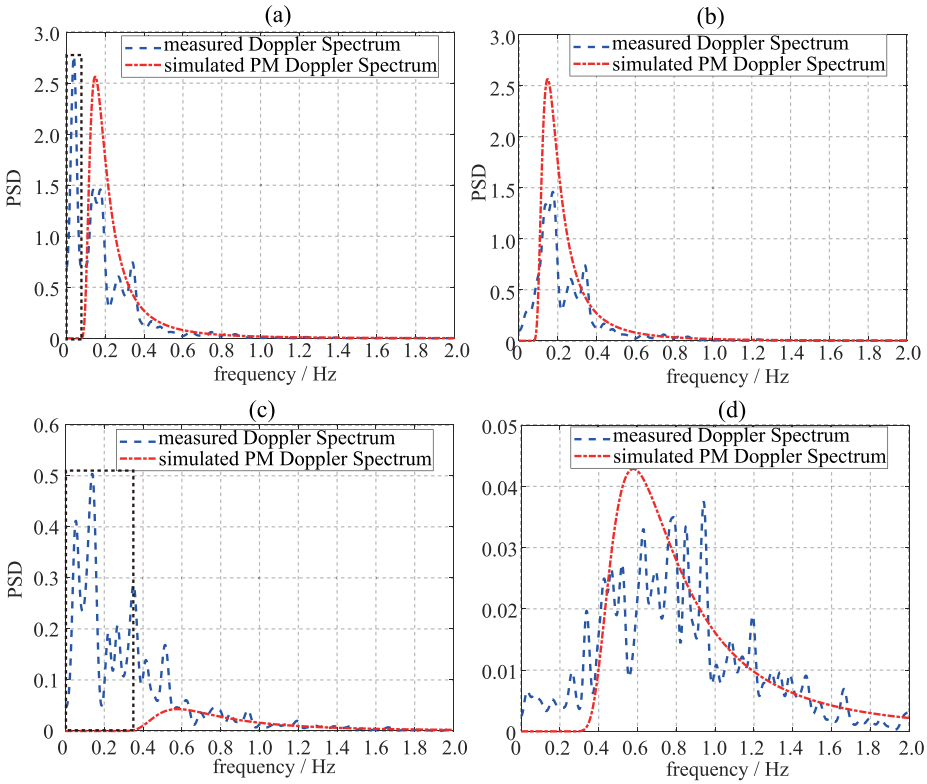


Fig. 6. Doppler spectra of (a) and (c) original micro-Doppler series and (b) and (d) reconstructed micro-Doppler series at the wind speed of (a) and (b) 8.6 m/s and (c) and (d) 2.1 m/s.

where $c_1(t)$ to $c_N(t)$ are the IMFs ordered from highest to lowest frequency, and $r(t)$ is the residual component. The micro-Doppler series due to wave motion is reconstructed by integrating only the high-frequency IMFs as

$$\widetilde{f}(t) = \sum_{i \in M(f_{\max})} c_i(t) \tag{43}$$

where the index set $M(f_{\max})$ is defined based on the peak frequency $f_{\max i}$ of each IMF.

$$M(f_{\max i}) = \left\{ f_{\max i} : f_{\max i} > D_f \right\} \tag{44}$$

where D_f is the given frequency threshold to separate the wave-motion spectrum from low-frequency interference, which empirically corresponds to the peak frequency of the Pierson–Moskwitz spectrum as [23]

$$D_f = 0.13 \frac{g}{u_{10}} \tag{45}$$

where u_{10} is the sea surface wind speed. The Doppler spectrum is then re-estimated from $\widetilde{f}(t)$ using Welch's method. Fig. 6(b) and (d) presents the Doppler spectra of the reconstructed micro-Doppler series for wind speeds of 8.6 and 2.1 m/s, respectively. The reconstruction effectively suppresses the low-frequency components, yielding spectra whose dominant frequency distributions align more closely with the theoretical simulations.

5) Estimation of Wind Speed:

An independent estimate of wind speed is obtained by fitting a theoretical model to the overall power spectrum of the reflected signal. The spectrum of the reflected GNSS signal is a Gaussian function of the correlation time and mean square height of the sea surface as

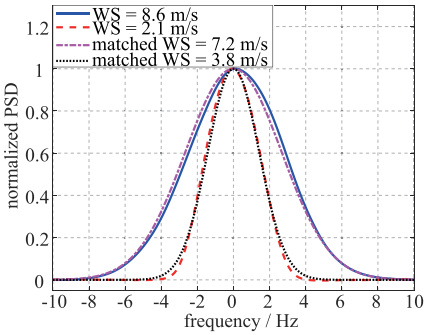


Fig. 7. Spectra of the reflected GNSS signals and the optimally matched theoretical spectra at the wind speed of 8.6 and 2.1 m/s.

$$S_{\text{spec}}(f) \sim \exp\left(-\frac{\pi^2 \tau_c^2 f^2}{2 k_s^2 \sin^2\theta \sigma_h^2} \right) \tag{46}$$

where k_s is the wavenumber of the GNSS signal, τ_c and σ_h are the correlation time and mean square height of the sea surface, computed from the wave spectrum model. The power spectral density of the reflected signal is obtained by integrating the time-Doppler spectrogram along the time axis

$$S_{\text{mea}}(f) = \frac{1}{T} \int_0^T S(t, f) dt \tag{47}$$

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where $S(t, f)$ is the time-Doppler spectrum. Fig. 7 demonstrates the agreement between the measured spectra and the optimally fit Gaussian models for wind speed of 8.6 and 2.1 m/s, respectively. The estimated spectrum align well with theoretical models. Wind speed is estimated by minimizing the discrepancies between the measured and theoretical spectral widths over all visible satellites

$$\hat{u}_{10} = \arg\min_{u_{10}} \left\{ \frac{1}{M_{\text{sat}}} \sum_{j=1}^{M_{\text{sat}}} \left(B_{\text{S}j} - \frac{2\sin\theta_j \sigma_h}{\lambda \tau_c} \right)^2 \right\} \tag{48}$$

where M_{sat} is the number of visible satellite. Sea waves consist of wind-driven waves and swell, which can affect wind speed retrieval. Under low wind speed conditions, the presence of dominant swell leads to an overestimation of the retrieved wind speed. Furthermore, under typhoon conditions, signal saturation and the nonequilibrium process between wind and wave may cause a systematic underestimation of the retrieved wind speed.

6) Estimation of Wind Direction:

Wind direction is retrieved using the mean micro-Doppler frequency shift [see (30)]. The retrieval approach is similar to that employed for wind speed, namely by minimizing the discrepancies between the measured and theoretical mean frequency shift over all visible satellites as

$$\widehat{\varphi} = \arg\min_{(\alpha,\varphi)} \left\{ \frac{1}{M_{\text{sat}}} \sum_{j=1}^{M_{\text{sat}}} \left(\alpha\cos(\varphi) - \frac{\overline{f}_{dj}}{B_{\text{S}j}} \right)^2 \right\}. \tag{49}$$

As indicated by (36) and Fig. 4, the mean frequency shift depends not only on wind direction but also on wind speed and elevation angle. Its amplitude increases with wind speed and varies linearly with the sine of the elevation angle. These variations align with the behavior of the spectral width under variations in wind speed and elevation angle. Consequently, to mitigate the influences of these factors, the mean frequency shift is normalized by the corresponding spectral width. It is noted that sea waves also significantly influence wind direction retrieval, and this influence varies with sea state. Under wind-wave-dominated conditions, the wind direction is aligned with wave propagation direction, favoring wind direction retrieval. In a swell-dominant sea, the wave direction may deviate completely from the wind direction, thereby misleading wind direction retrieval.

7) Synthesis of Doppler Spectrum:

Multiple GNSS satellites are typically visible simultaneously, allowing for a reduction of estimation variance by averaging Doppler spectra from all available satellites. Since the Doppler spectrum depends on the elevation angle of the GNSS satellite [see (22)], a calibration is applied before averaging. The synthesized Doppler spectrum is computed as

$$\overline{F}(f) = \frac{1}{M_{\text{sat}}} \sum_{j=1}^{M_{\text{sat}}} \frac{\widetilde{F}_j(f)}{\sin^2\theta_j} \tag{50}$$

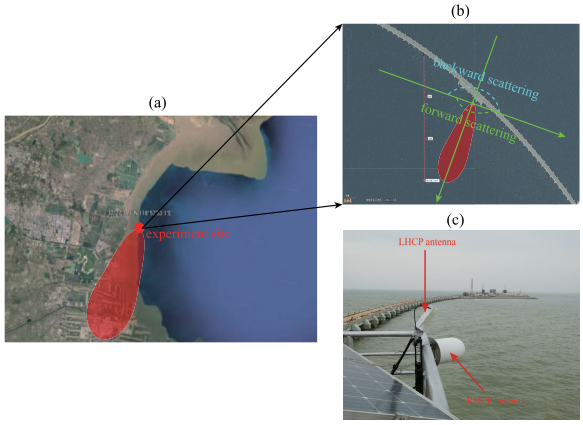


Fig. 8. Scenario of the coastal GPS L1 and BeiDou B1I signal experiment. (a) Experiment site on map, (b) boresight direction of the sea-looking antenna's mainlobe projected on a local map, and (c) antenna setup.

where $\widetilde{F}_j(f)$ and θ_j are the Doppler spectrum and the elevation angle of the jth GNSS satellite; and M is the number of the visible GNSS satellites.

8) Estimations of Micro-Doppler Characteristics and Inversion of Wave Spectrum:

The final synthesized Doppler spectrum is used to compute the micro-Doppler characteristics via (23)~(26). The micro-Doppler amplitude f_{fmax} and f_{fequ} are obtained from the reconstructed time series $\widetilde{f}(t)$ using (27) and an optimal Gaussian matching method, respectively. According to (22), the wave spectrum is retrieved as

$$\hat{S}(f) = \frac{\overline{F}(f)}{TF} \tag{51}$$

where TF represents the transfer function as $32\pi^2 f_\Delta f^2 / \lambda^2$.

9) Estimations of Wave Parameters:

The wave height spectrum can be retrieved from the synthesized Doppler spectrum $\overline{F}(f)$ through (50) and (51). Once the wave height spectrum $\hat{S}(f)$ is obtained, key wave parameters, including SWH H_s and mean wave period T_{ave}, are calculated as

$$H_s = 4\sqrt{\int_0^{+\infty} \widehat{S}(f) df} \tag{52}$$

$$T_{\text{ave}} = \frac{\int_0^{+\infty} \widehat{S}(f)\, df}{\int_0^{+\infty} f\widehat{S}(f)\, df}. \tag{53}$$

Before the integration, a high-pass filter with a cutoff of f_{cutoff} is used in the estimated sea wave spectrum to mitigate the low-frequency interference. In this article, f_{cutoff} is set as $f_{\text{cutoff}} = D_f$.

IV. Experimental Demonstrations and Analysis

A. Experimental Scenarios

As depicted in Fig. 8(a), the experiment was conducted in Dongying, Shandong Province (37°26′51″N, 119°0′36″E). The campaign consisted of two stages: the first from October

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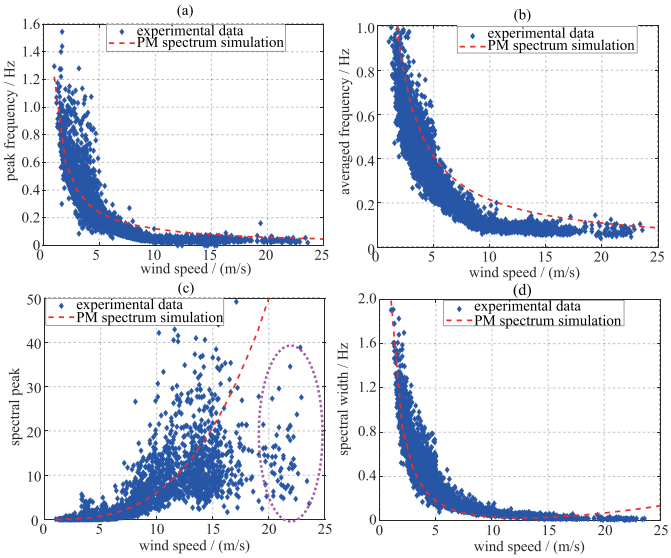


Fig. 9. Dependencies of (a) peak frequency, (b) averaged frequency, (c) spectral peak, and (d) spectral width of the Doppler spectrum on wind speed.

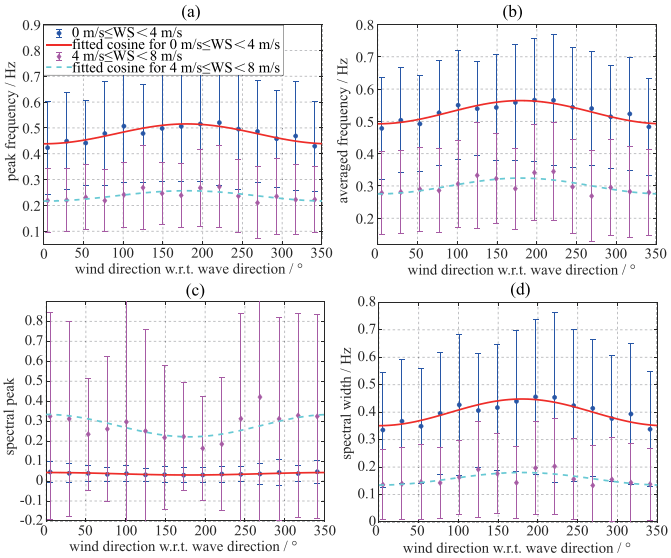


Fig. 10. Dependencies of (a) peak frequency, (b) averaged frequency, (c) spectral peak, and (d) spectral width of the Doppler spectrum on the included angle between wind and wave directions.

26 to November 20, 2021, and the second from August 27 to October 15, 2025. A coastal GNSS-R receiver, designed and developed as described in [11], was employed to simultaneously process direct and reflected GPS L1 and BeiDou B1I signals in real-time, outputting the corresponding complex correlation values. The antenna configuration is illustrated in Fig. 8(b). Two receiving antennas were mounted approximately 12 m above mean sea surface level. The antenna dedicated to the reflected signal was a four-element array with a peak gain of 9.4 dBi, a mainlobe width of 60°, and was oriented at an azimuth angle of 210°. The red lobe in Fig. 8 denotes the boresight direction of this antenna's mainlobe projected on a map. For ground-truth validation, in situ wind speed and direction were obtained from a cup anemometer installed near the experiment site. The SWHs were provided by a nearby buoy. Reference values for mean wave period and direction were derived by spatially and temporally interpolating the fifth-generation European Centre for Medium-Range Weather Forecasts (ERA5) dataset.

B. Spectrum Analysis

1) Wind Speed: Fig. 9 illustrates the relationships between the four Doppler spectrum characteristics and wind speed. As wind speed increases, both the peak frequency f_{p} and mean frequency f_{a} exhibit a gradual decrease. This trend aligns with the established phenomenon that the dominant wave energy shifts toward lower frequencies under higher wind speeds. The Doppler spectrum peak S_{peak} shows a clear increasing variation with wind speed, while the spectral width B_S generally decreases. At low wind speeds, wave energy is more dispersed across frequencies, resulting in a lower and broader spectral peak. As wind speed increases, waves grow, and their energy becomes more concentrated near the dominant frequency. This leads to a higher, sharper spectral peak, which is a behavior consistently observed in ocean wave spectra and now reflected in the derived Doppler spectra. The theoretical relationships, derived from the Pierson–Moskwitz wave spectrum, are plotted as the dashed curves in Fig. 9. The close agreement between the measured data and the model predictions demonstrates the validity of the theoretical model [see (22)] for the Doppler spectrum and confirms the feasibility of extracting the wave spectrum from GNSS-R observations.

TABLE I

BEST-FIT SPREADING PARAMETERS FOR THE FOUR CHARACTERISTICS OF THE DOPPLER SPECTRUM

	$0 \le u_{10} < 4$ m/s	$4 \le u_{10} < 8$ m/s
peak frequency	1.81	0.88
averaged frequency	1.73	0.95
spectral peak	1.37	0.33
spectral width	1.53	1.09

2) Wind and Wave Direction: The directional distribution of wave energy is a fundamental property of sea waves, which describes how the energy is distributed across different propagation directions (referred to as wind direction) at each frequency. As indicated in (22), the Doppler spectrum inherits the directional modulation function. Under the common $\cos^{2s}$-type spreading model, a cosine-shape variation is expected in the Doppler spectrum characteristics with respect to φ. Fig. 10 presents the dependence of the four spectral characteristics on the included angle between wind and wave directions. All characteristics exhibit variations consistent with a cosine modulation. When the dominant wave direction aligns with wind direction ($\varphi = 0°$), the spectral peak is maximized, indicating that wave energy is most concentrated in the downwind direction. Conversely, the peak frequency, mean frequency, and spectral width show the opposite trends, being minimized near $\varphi = 0°$. This frequency-dependent behavior suggests that the spreading parameter s in the direction spreading function is a function of frequency, attaining its maximum value near the peak frequency. When wind and wave directions are aligned, energy input is more efficient, promoting a fuller development

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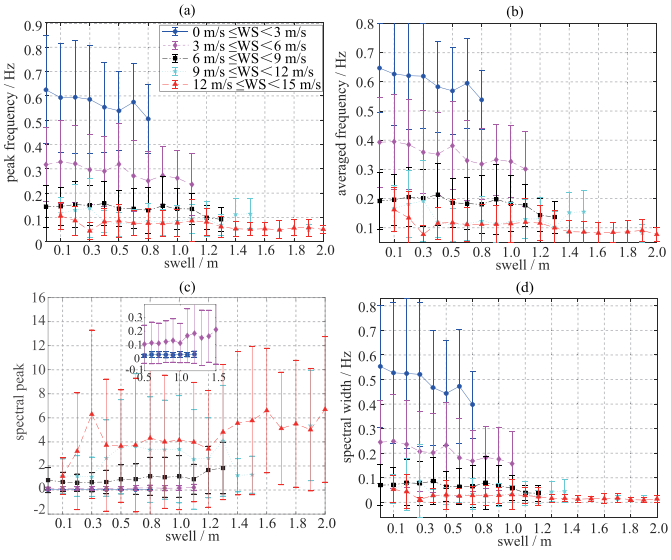


Fig. 11. Dependencies of (a) peak frequency, (b) averaged frequency, (c) spectral peak, and (d) spectral width of the Doppler spectrum on swell. The zoomed-in-view plot in (c) presents the variations of the spectrum peak with swell from 0.5 to 1.5 m for wind speed ranges of 0–3 m/s and 3–6 m/s, respectively.

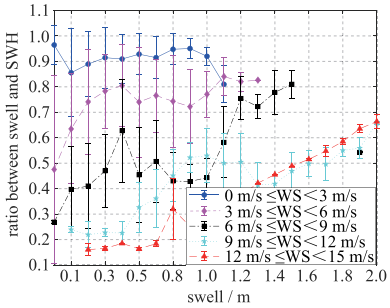


Fig. 12. Ratio of swell SWH to total SWH at different swell heights.

of the spectrum toward lower frequencies, and resulting in a narrower and higher peak and more concentrated energy distribution. To demonstrate that the spreading parameter s varies with ocean conditions, best-fit spreading parameters s are estimated for two wind speed intervals, and summarized in Table I. The values are generally lower for the moderate wind speed interval of $[4, 8)$ m/s compared to the low wind speed interval of $[0, 4)$ m/s. Wind speed and swell are two primary external forces shaping the directional wave spectrum, with distinct mechanisms and effects. At very low wind speeds, remotely generated swell often dominates the wave field, exerting a higher influence on the spreading parameter than wind speed. Swell is typically more directionally pure (narrowly distributed), which explains the higher effective spreading parameters fit in that regime.

3) Swell Wave: Swell waves, characterized by their long periods and smooth crests, originate from distant storms and dominate the low-frequency portion of the wave spectrum. Previous studies have demonstrated a significant influence of swell on reflected GNSS signals, particularly at low wind speeds (<6 m/s) [24]. Fig. 11 shows the dependencies of the Doppler spectrum characteristics on swell SWH. The

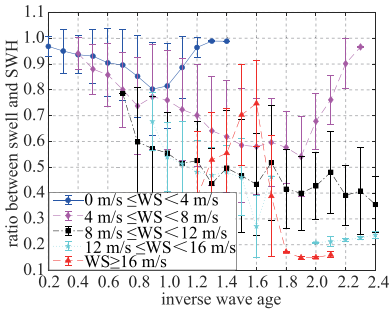


Fig. 13. Ratio of swell SWH to total SWH at different inverse wave ages.

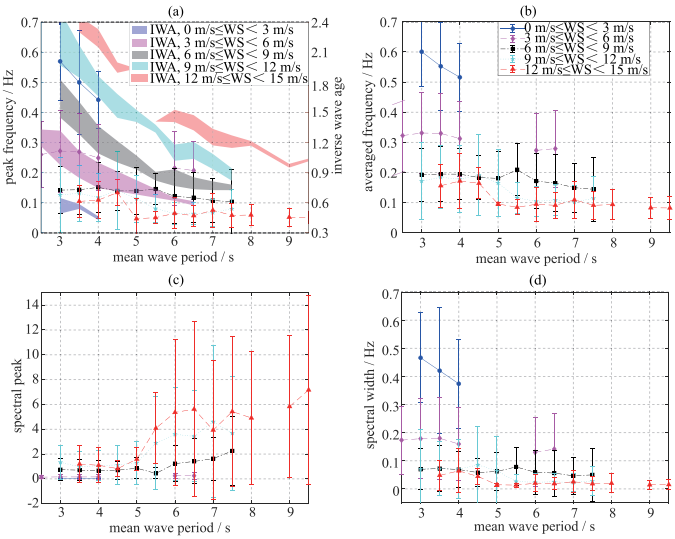


Fig. 14. Dependencies of (a) peak frequency and inverse wave age, (b) averaged frequency, (c) spectral peak, and (d) spectral width of the Doppler spectrum on the dominant wave period.

influences of swell pronounced at wind speeds below 6 m/s, consistent with existing works. As illustrated in Fig. 12, under these conditions, swell can constitute over 70% of the total wave energy, causing the wave spectrums to assume swell-like properties. With increasing swell SWH, the peak frequency, mean frequency, and spectral width tend to decrease, while the spectral peak increases. This pattern reflects that the statistical properties of swell: larger swell generally have longer periods and carry more concentrated, higher energy at lower frequencies.

4) Wave Period or Inverse Wave Age: The inverse wave age is a key parameter describing the developmental stage of wind-generated waves and indicates the energetic relationship between the local wind and wave field. It is defined as

$$\omega_c = 2\pi \frac{u_{10}}{gT} \tag{54}$$

where T is the mean wave period. Low inverse wave ages indicate a swell-dominated regime, while high values signify a wind-driven wave regime. This is corroborated in Fig. 13, which shows that swell contributes over 80% of the total SWH when the inverse wave age is less than 0.8. Once the inverse wave age exceeds 0.8, the proportion of swell decreases, and wind-driven waves dominate the total waves.

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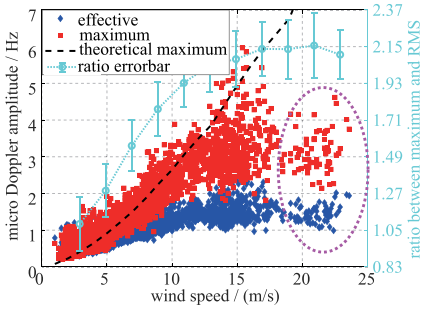


Fig. 15. Relationships between micro-Doppler amplitude and wind speed, and corresponding ratio error bar of the maximum amplitude to the effective (rms) amplitude.

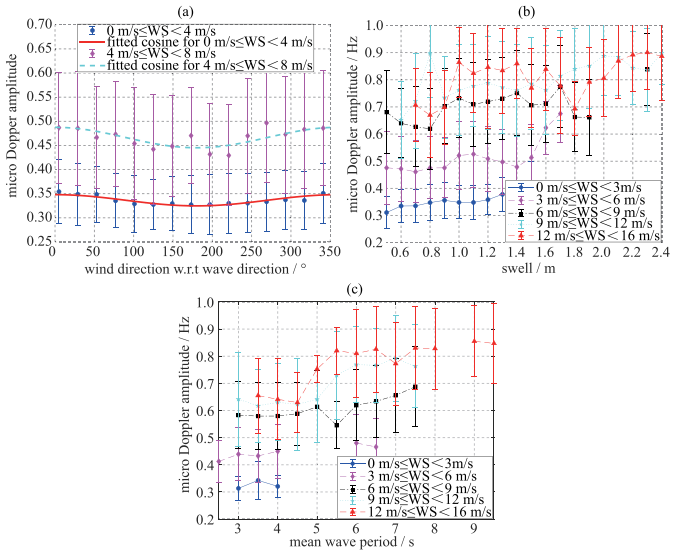


Fig. 16. Dependencies of micro-Doppler amplitude on the (a) included angle between wind and wave direction, (b) swell, and (c) dominant wave period.

For a given wind speed, the inverse wave age is inversely proportional to the corresponding mean wave period [see (54)]. This relationship also illustrated in Fig. 14(a). The clear dependencies between the Doppler spectrum characteristics and the mean wave period are demonstrated in Fig. 14. As mean wave period increases (or the inverse wave age decreases), signifying a more developed or swell-influenced sea state, the peak frequency, mean frequency, and spectral width decrease. Conversely, the spectral peak increases. This shift indicates the transition of dominant energy to lower frequencies.

C. Micro-Doppler Amplitude Analysis

As theoretically established by (28), the micro-Doppler amplitude is governed by the spectral peak of the wave height spectrum. Fig. 15 presents the measured maximum and Gaussian-fit (equivalent) amplitudes as a function of wind speed. Both amplitude metrics show a general increase with wind speed, and the trend of the maximum amplitude aligns well with the theoretical prediction from (28). Saturation of the measured amplitudes is observed at wind speeds exceeding 18 m/s, which also reflected in the behavior of the Doppler spectral peak shown in Fig. 9(c). In addition, the ratio of the

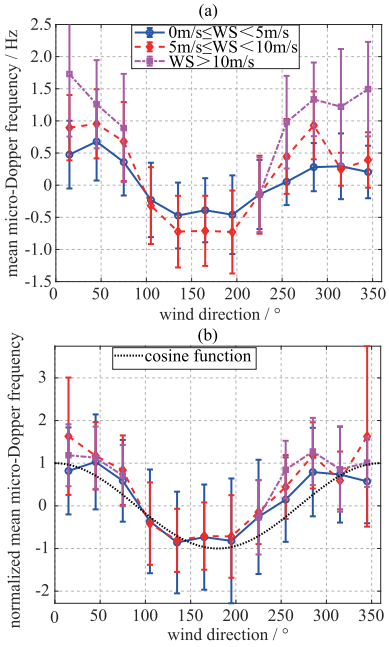


Fig. 17. (a) Mean micro-Doppler frequency shifts versus wind direction for different wind speed intervals and (b) corresponding normalized mean micro-Doppler frequency shifts versus wind direction.

maximum amplitude to the effective (rms) amplitude increases with wind speed, indicating that the micro-Doppler frequency shift becomes more random at higher wind speeds. The dependencies of the micro-Doppler amplitude on the relative angle between wind and wave direction, swell SWH, and dominant wave period are illustrated in Fig. 16. The observed influences of these parameters on the amplitude are consistent with their effects on the Doppler spectral peak, as detailed in (28) and Figs. 10(c), 11(c), and 14(c). This agreement provides further experimental validation for the theoretical model of micro-Doppler amplitude presented in (28).

D. Mean Micro-Doppler Frequency Shift Analysis

Fig. 17(a) presents the dependencies of the estimated mean micro-Doppler frequency shift on wind direction across different wind speed intervals. The mean frequency shift exhibits a symmetrical variation with wind direction, with symmetry axis of about 180°. Additionally, the shift amplitude increases with increasing wind speed. These phenomena align with the simulated results in Fig. 4. To reduce the influence of wind speed and elevation angle, the mean frequency shift is normalized by the corresponding spectral width. Fig. 17(b) displays the variations of the normalized frequency shift $\lambda\overline{f}_d/B_S \sin\Delta\theta$ ($\Delta\theta = 2°$) with wind direction. The normalized frequency shifts follows a consistent cosine-like pattern across wind speed intervals. These findings indicate that, in addition to the specular reflection, which is primarily sensitive to the mean-square slope, and weakly modulated by direction, a distinct off-specular scattering is present in the GNSS-R forward scattering geometry. This scattering shows clear sensitivity to directional modulation.

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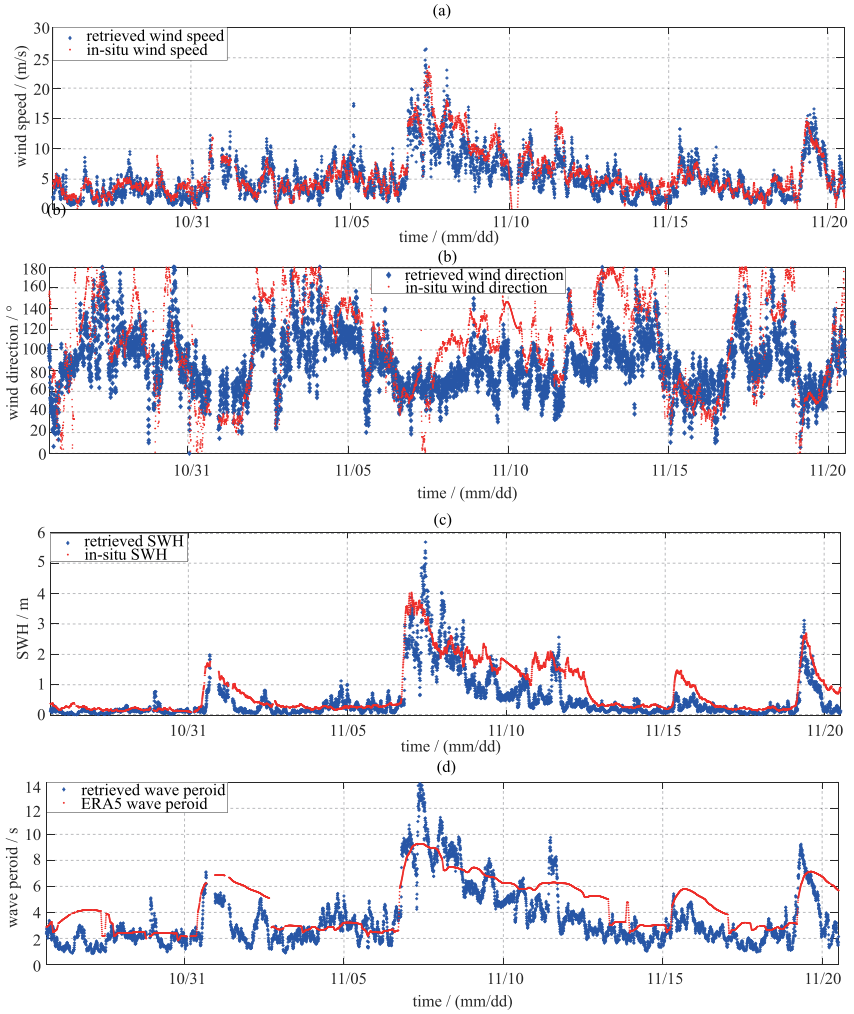


Fig. 18. Retrieved (a) wind speed, (b) SWH, (c) wave period, and corresponding referenced measurements in the first-stage experiment.

V. Application in Sea State Observation

The retrieval performance for wind speed, wind direction, SWH, and mean wave period is evaluated against the reference data during both experimental stages. Wind speed is retrieved from the integrated time-Doppler spectrogram using (48). Wind direction is derived from the mean micro-Doppler frequency shift via (49). SWH and mean wave period are estimated from the derived wave height spectrum using (52) and (53), respectively. The retrieved results and corresponding comparisons are presented in Figs. 18 and 19 (with $\Delta f = 0$). The retrieved wind speeds via (48) exhibit close agreement with in situ wind speeds, yielding root mean square errors (RMSEs) of 2.55 and 1.74 m/s for the first- and second-stage experiments, respectively. A maximum retrieved wind speed close to 25 m/s, attributed to a local cyclone on November 17, 2021, demonstrates the method's capability to capture high wind speed in coastal regions. The retrieved wind directions during the first- and second-stage experiments yield RMSEs of 37.04° and 32.09°, respectively. Larger errors occur primarily in cases with notable underestimations, owing to the influence of wave direction. For SWH, the RMSEs are 0.56 and 0.55 m in the first- and second-stage experiments, respectively. The corresponding RMSEs for the retrieved wave periods are 1.77 and 1.54 s, respectively. Notably, the mean wave period used for comparison is derived from the ERA5 reanalysis dataset, which has an hourly temporal resolution and a $0.5° \times 0.5°$ spatial grid. ERA5 is inherently limited in resolving the high-frequency temporal and fine spatial variability of coastal waves. This deficiency is also evident from the smooth variations in Figs. 18(c) and 19(c). Consequently, the comparison presented here primarily demonstrates the feasibility of retrieving the mean wave period using the reflected GNSS signal, rather than characterizing the definitive retrieval accuracy achievable under ideal reference conditions. In the experimental setup, the sea-looking antenna used had a beam gain of 9.4 dBi, which is not optimal for receiving the weak reflected GNSS signal. Utilizing an antenna with higher gain would enhance the signal-to-noise ratio and is expected to improve the retrieval performance.

VI. Discussion

As shown in Fig. 14, inverse wave age exhibits an important influence on the sea wave spectrum. For an inverse wave age below 0.4, as shown in Fig. 20, the presence of swell

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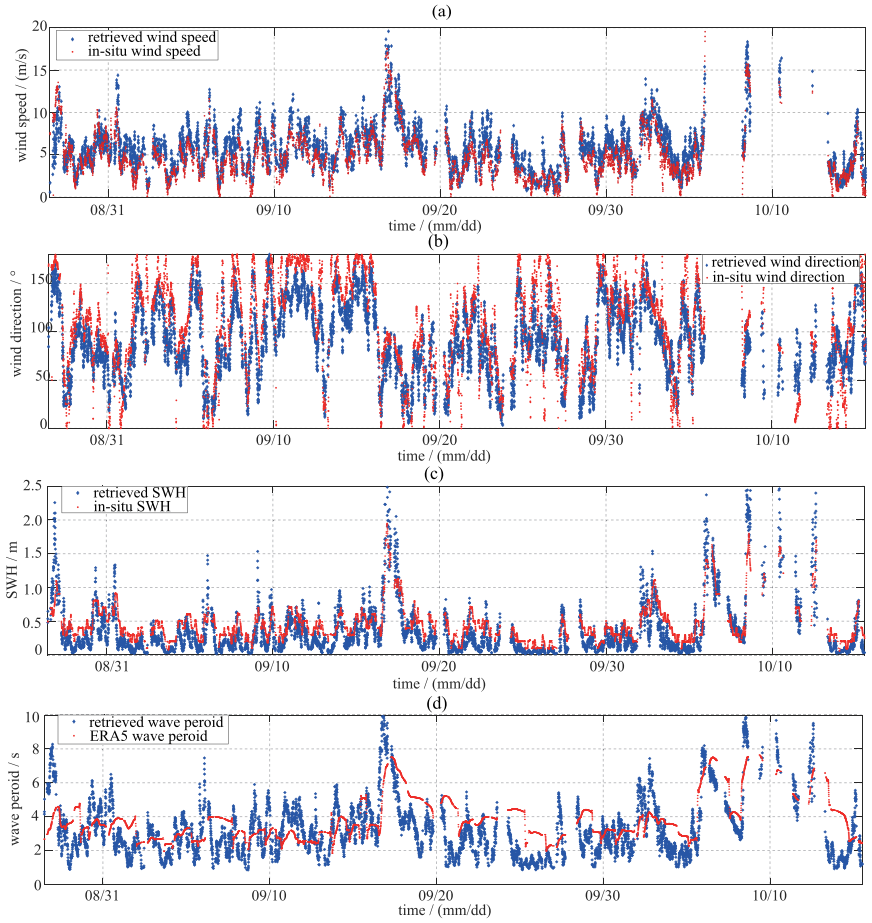


Fig. 19. Retrieved (a) wind speed, (b) wind direction, (c) SWH, (d) wave period, and corresponding reference measurements in the first-stage experiment.

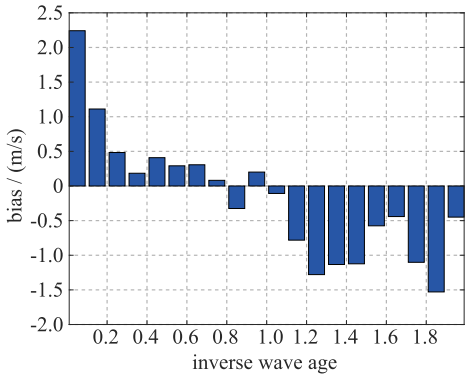


Fig. 20. Biases of the retrieved wind speed at different inverse wave ages.

leads to an overestimation of the retrieved wind speed. For an inverse wave age exceeding 1.0, and for a given wind speed, a higher inverse wave age results in a lower spectral peak, a higher peak frequency, and a wider spectral width of the sea wave spectrum. This change indicates that when using the simulated observable derived from the sea wave spectrum while ignoring inverse wave age to fit the estimated observable for wind speed retrieval, an underestimation may occur in cases of higher inverse wave age. A higher inverse wave age typically occurs at higher wind speeds, indicating that the retrieved wind speed using (48) may be underestimated at high wind speeds. To address this underestimation, an inverse-wave-age parameterized sea wave spectrum, such as the Elfouhaily spectrum [25], can be adopted. The best-fitting approach between the inverse-wave-age parameterized sea wave spectrum and the estimated spectrum can be employed to jointly retrieve the mean wave period and wind speed. Unfortunately, the estimated sea wave spectrum is distorted by some low-frequency interferences, resulting in imprecise retrieval of wind speed and mean wave period. Therefore, novel algorithms should be explored to mitigate these low-frequency interferences. In addition, the signal-to-noise ratio (SNR) is an important indicator of signal quality. As shown in Fig. 21, a higher SNR is beneficial for improving of retrieval precision.

As shown in Fig. 22, when waves approach the coast, the retrieved wind direction has a larger RMSE; correspondingly, offshore waves produce a relatively lower RMSE for the retrieved wind direction. This result indicates that wave direction has an important impact on the retrieval of wind direction, thereby limiting the precision of the retrieved wind direction. In addition, the retrieved wind direction exhibits a 180° ambiguity, caused by the directional symmetry of the sea wave spectrum. This ambiguity can be removed by multibeam

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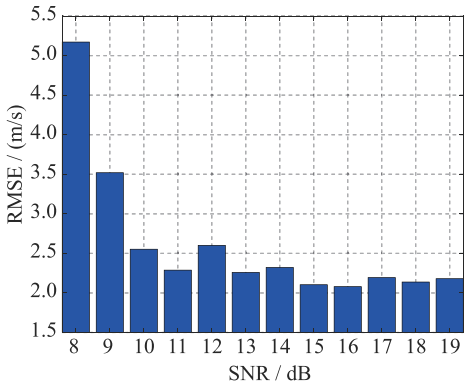


Fig. 21. RMSEs of the retrieved wind speed at different SNR levels.

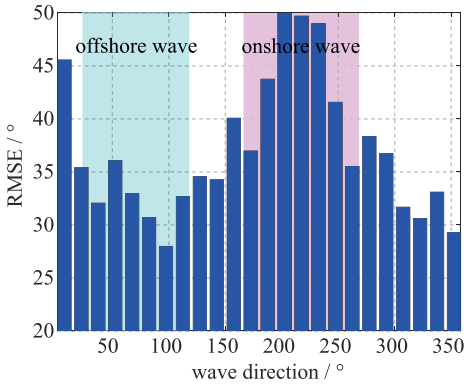


Fig. 22. RMSEs of the retrieved wind direction at different wave directions.

observation [26]. Consequently, multiple correlators for processing the reflected GNSS signals received by distinct beams are needed in the coastal GNSS-R receiver.

VII. Conclusion

Conventional monostatic radar echoes are modulated by ocean wave motion, enabling the retrievals of wave motion parameters, such as SWH and mean wave period. In contrast, GNSS-R operates as a bistatic radar, utilizing specular-reflected GNSS signals to retrieve oceanic physical parameters. The prevailing physical mechanism for sea state retrieval (e.g., wind speed) in GNSS-R relies on signal decorrelation caused by sea surface roughness. However, due to the insensitivity of specular reflection to horizontal motion, the explicit modulation of reflected signals by wave orbital kinematics has received limited attention. Crucially, waves possess a vertical orbital velocity component, which induces a weak, time-varying oscillation in the Doppler frequency of the reflected signal, that is termed the micro-Doppler effect in this study to distinguish it from the bulk Doppler shift caused by satellite motion. Beyond wave motion, other phenomena such as wave breaking introduce low-frequency interference. To address this, a novel processing method combining dynamic programming and EMD was developed to extract the wave-induced micro-Doppler component. From the derived micro-Doppler time series, a corresponding Doppler spectrum was obtained. Both theoretical analysis and experimental validation confirmed that this Doppler spectrum is fundamentally related to the wave height spectrum. To quantify the micro-Doppler's dependence on sea state, several characteristic metrics were defined and analyzed against parameters including wind speed, wind direction, wave direction, swell, and mean wave period. In particular, the mean micro-Doppler frequency shift was theoretically derived under the assumption that off-specular scattering occurs in GNSS-R forward observations. This nonzero mean micro-Doppler frequency shift exhibits a symmetrical dependence on wind direction about a 180° axis. This property provides a practical scheme for retrieving wind direction, thereby overcoming a fundamental limitation of coastal GNSS-R, which was previously considered incapable of retrieving wind direction. The experimental relationships align well with theoretical predictions, thereby demonstrating the modulation of reflected GNSS signals by wave orbital motion and the presence of the off-specular scattering, and establishing the feasibility of wind direction, wave spectrum, and wave parameters sensing via GNSS-R. The retrieved wind direction using the mean micro-Doppler frequency shift achieved an RMSE of 32°. Using the retrieved wave height spectra, SWH and mean wave period were estimated, achieving root-mean-square errors (RMSEs) of approximately 0.50 m and 1.50 s, respectively.

While performance may vary under more complex sea states, this work provides a foundational demonstration of GNSS-R's capability to observe wave motion and retrieve wave spectra. It also motivates further methodological advances in the remote sensing of wave kinematics. Moreover, the proposed framework is not limited to GNSS signals; it is equally applicable to other opportunistic signal sources, such as those from communication satellites, broadening the potential for multisource ocean monitoring. In the experiment reported in this article, the sea-looking antenna was a directional antenna with a peak gain of 9.4 dB. Such a higher-gain antenna is beneficial to capture fine-scale wave motion.

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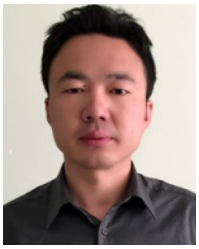
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